University of Denver
Digital Commons @ DU

Undergraduate Theses, Capstones, and Recitals | Undergraduate Research

Spring 6-14-2025

Urban-Rural Divide in Inequality's Impact on Substance Abuse

Jonathan E. Flores

Follow this and additional works at: https://digitalcommons.du.edu/undergraduate_theses

Part of the Economic Policy Commons, Health Economics Commons, Inequality and Stratification Commons, Public Health Commons, Substance Abuse and Addiction Commons, and the Urban Studies and Planning Commons

All Rights Reserved.

Urban-Rural Divide in Inequality's Impact on Substance Abuse

Abstract

This thesis focuses on investigating the relationship between income inequality, measured through the Gini coefficient, and drug addiction, proxied through substance-related mortality. Specifically, this thesis focuses on how these dynamics vary across rural and urban counties in the United States, using population size as a proxy. County-level data from 2016 to 2020 were collected through both the CDC and the Census Bureau and subsequently analyzed using Stata, with crude death rates from drug and alcohol causes serving as a function of the Gini coefficient, labor force participation, racial composition, and other socioeconomic indicators. Through a quadratic version of the model, a U-shaped relationship between income inequality and drug-related deaths was revealed, suggesting that both extremely equal and highly unequal counties face elevated risk (albeit for different structural reasons). Labor force participation also exhibited a nonlinear but predominantly negative association, with some evidence of an income effect at higher participation levels. Counties with higher populations of white individuals were also a significant predictor of drug mortality, aligning with the literature on "deaths of despair" in America. There were some different patterns between drug crude rates and alcohol crude rates. Ultimately, these findings express that interventions (especially economic ones) must take a nuanced approach to addressing substance abuse at the county level.

Document Type

Undergraduate Honors Thesis

Degree Name

B.A. in Economics

First Advisor

Juan Carlos Lopez

Keywords

Income inequality, Drug mortality, Crude death rate, Gini coefficient, Labor force participation, Nonlinear regression, Rural–urban divide, Deaths of despair, Substance use, County-level analysis, Opioid epidemic, Public health disparities

Subject Categories

Economic Policy | Economics | Health Economics | Inequality and Stratification | Public Health | Substance Abuse and Addiction | Urban Studies and Planning

Publication Statement

Copyright is held by the author. User is responsible for all copyright compliance.

The University of Denver

Urban-Rural Divide in Inequality's Impact on Substance Abuse

Thesis Submitted to

The Economics Department

And Faculty of Honors Programs

For Distinction in a Bachelor of Arts in Economics

College of Arts Humanities and Social Sciences

By

Jonathan E. Flores

Advisor: Dr. Juan Carlos Lopez

Denver, Colorado

June 2025

Abstract

This thesis focuses on investigating the relationship between income inequality, measured through the Gini coefficient, and drug addiction, proxied through substance-related mortality. Specifically, this thesis focuses on how these dynamics vary across rural and urban counties in the United States, using population size as a proxy. County-level data from 2016 to 2020 were collected through both the CDC and the Census Bureau and subsequently analyzed using Stata, with crude death rates from drug and alcohol causes serving as a function of the Gini coefficient, labor force participation, racial composition, and other socioeconomic indicators. Through a quadratic version of the model, a U-shaped relationship between income inequality and drug-related deaths was revealed, suggesting that both extremely equal and highly unequal counties face elevated risk (albeit for different structural reasons). Labor force participation also exhibited a nonlinear but predominantly negative association, with some evidence of an income effect at higher participation levels. Counties with higher populations of white individuals were also a significant predictor of drug mortality, aligning with the literature on "deaths of despair" in America. There were some different patterns between drug crude rates and alcohol crude rates. Ultimately, these findings express that interventions (especially economic ones) must take a nuanced approach to addressing substance abuse at the county level.

Contents

Literature Review…… 1

Framing the Crisis: Addiction, Inequality, and Deaths of Despair……. 1

Theoretical Foundations: Rational Addiction, Social Stress, and Deaths of Despair……. 2

Economic Inequality and Substance Use…… 3

The Rural–Urban Divide in Addiction and Inequality ……... 8

Gaps and Rationale for the Present Study……. 11

Methods…… 13

Data Sources and Collection……. 13

Data Integration and Preparation…….. 14

Empirical Strategy and Statistical Modeling…… 14

Results…….. 15

Summary Statistics……... 15

Regression Results: Drug Crude Rate Mortality…….. 17

Regression Results: Alcohol Crude Rate Mortality…….. 23

Discussion…… 25

Drug Crude Rates and the White Population…… 26

Nonlinear Effects of Income Inequality…… 26

Interpreting Labor Force Participation Trends…….. 27

Conclusion…… 28

Limitations……...……. 28

Future Research……...……. 30

References……...…… 31

Appendix…….…… 34

Literature Review

Framing the Crisis: Addiction, Inequality, and Deaths of Despair

The United States is in the midst of a devastating drug addiction and overdose epidemic. Opioid-related overdose deaths have skyrocketed in the past two decades, reaching approximately 47,590 deaths in 2018, which is a threefold increase from only a decade earlier (Fischer et al., 2020). This surge in fatalities contributed to the recent declines in life expectancy around that time. In essence, what began as a prescription opioid misuse from the 1990s to the 2000s has transformed into a crisis, increasingly driven by potent synthetic opioids like fentanyl in the 2010s (Marks et al., 2021). Communities across the country, from rural Appalachia all the way to the dense urban centers, have borne a heavy burden. Many scholars have framed this public health emergency as part of a broader trend known as "deaths of despair." This term was coined by the economists Anne Case and Angus Deaton, who essentially refer to the mortality from suicide, drug overdose, and alcohol-related disease that has risen among demographics, specifically White Americans without college degrees, facing economic and social decline (Scutchfield & Keck, 2017). This phenomenon is unique to the U.S., as other high-income countries see falling mortality rates with their middle-aged population. Case and Deaton attribute these American trends to factors such as worsening job prospects (which include stagnant or declining wages), the loss of stable institutions (e.g., declining marriage and religious participation), and the eroding social supports. Ultimately, this concoction of economic and social hardships generates hopelessness, which makes individuals prone to substance abuse and other self-destructive behaviors.

Additionally, there is interest among researchers and policymakers regarding how income inequality may be fueling drug addiction and overdose-related deaths, and whether this plays out differently in rural vs urban areas. Income inequality in the U.S. has risen sharply in recent

decades, and many of the communities suffering the worst of the opioid crisis are also those marked by economic deprivation, which is connected to the widening gaps between the rich and poor. This literature review will explore existing research on the link between economic factors and substance use outcomes, which will be the center of this review.

Theoretical Foundations: Rational Addiction, Social Stress, and Deaths of Despair

To better understand drug addiction from the context of economics, it requires multiple different perspectives. Three frameworks of interest into why economic and social conditions may drive substance use include (but are not limited to) Rational Addiction Theory, Social Stress and Coping Models, and the Death of Despair frameworks. It is important to note that there is an overlap with all three frameworks, so no one is superior to the other, but they each provide a perspective and guidance of their own.

Gary Becker and Kevin Murphy's Rational Addiction Theory provides a solid economic lens for modeling drug use. They describe it as a forward-looking, utility-maximizing behavior despite its potential long-term costs (Baltagi & Griffin, 2001). Based on this theory, individuals are assumed to weigh the current pleasurable utility of consuming an addictive substance against the future disutility (health problems, dependence, financial constraints, etc.) in a consistent, rational manner. Becker and colleagues argue that addiction can be understood from standard consumer choice models, where people will consume more of a drug if the discounted future harms are outweighed by the immediate benefits based on their preferences and constraints. What this model essentially implies is that the decision to use drugs is sensitive to one's expectations about the future. In contexts of inequality, those constraints and expectations may differ across socioeconomic groups. For example, an individual in a depressed community with limited economic opportunity might rationally (in an economic sense) place lower value on their future well-being because their prospects for improvement are dim. Instead, there will be a

higher value on the present relief of pleasure from their substance use. To summarize, extreme inequality and lack of upward mobility could shorten one's time horizon, effectively making the immediate escape of drugs "worth it" relative to the long-term benefit of abstinence. This pure rational model of action assumes that individuals have full information and consistent behavior, which garners its fair share of criticism, citing that addiction often involves imperfect information and self-control issues (Baltagi & Griffin 2001). Nevertheless, this framework is a useful starting point: it suggests that economic conditions alter the constraints under which individuals optimize their behavior.

Economic inequality is not only an issue of material resources, but also a source of psychosocial stress. Sociological and public health theories argue that living in a highly unequal environment, specifically for those on the lower levels, generates chronic stress, erodes social bonds, and can precipitate coping behaviors such as substance abuse. This framework is the Social Stress Model of substance abuse, which suggests that individuals facing high levels of stress (which can come from economic hardship) are more likely to engage in drug use, particularly if they lack adequate coping resources or social support (Wilkinson & Pickett, 2009). Wilkinson and Pickett's cross-national research argues that societies with large income gaps create more status competition and insecurity, which adversely affects mental health and social cohesion. Examples of these social problems include violence, mental illness, and drug abuse, which are more common in more unequal societies, likely because larger status differences intensify stress across the social hierarchy. Fundamentally, the stress of inequality can be seen as toxic, with those at the bottom experiencing shame and hopelessness, and even those in the middle feeling anxiety about falling behind.

There is also empirical evidence from animal studies that supports a stress-addiction link. In an experiment with primates, Gould et. al. examined the connection of social status and how that influences the appeal of drugs. Male macaque monkeys were placed in established social hierarchies and then subjected to stressful social confrontation by being introduced into another group. As a result, the subordinate low-status monkeys showed a significant increase in sensitivity to the reinforcing effects of cocaine after the stressful confrontation, whereas dominant monkeys showcased decreased sensitivity. Brain scans between the low-status and high-status monkeys indicate divergent patterns of brain metabolism consistent with differences in stress and anxiety processing. Overall, these findings demonstrate support for the social stress theories, which indicate that individuals with lower social status are more vulnerable to stress-induced increases in drug use. The social stress framework provides a psychosocial mechanism by which inequality may translate into addiction; drugs and alcohol become a form of self-medication for the stress and despair of one's socioeconomic circumstances.

Case and Deaton's Death of Despair framework offers a narrative that connects long-term socioeconomic decline to the increase of substance abuse, suicide, and related mortality. From this perspective, the opioid epidemic is not an isolated public health issue, but actually a symptom of deeper economic and social decay. This decay that Case and Deaton are referring to is that which has happened since the late 20th century, where white working-class Americans have experienced stagnating and falling economic fortune. This outcome is operationalized through the loss of well-paying manufacturing jobs, stagnating wages, and weakened social institutions (marriage, church, civic organizations, etc.). Consequently, there has been a loss of hope and identity, contributing to despair (Case & Deaton 2021). The defined deaths of despair are those that include suicide, drug overdoses, or alcoholic liver disease, which are "self-

inflicted" in the sense that they arise from behaviors used to cope (albeit maladaptively) with pain. This death of despair framework can be summarized as such: long-term economic deterioration, social breakdown, and an increase in substance abuse and suicide. This pattern has been prevalent in communities where good jobs have vanished, leaving no clear future for the upcoming generation.

Economic Inequality and Substance Use

Much research has been done on the relationship between income inequality and health outcomes, especially with cross-national comparisons. Wilkinson and Pickett (2009) showed that more unequal societies often suffer from higher rates of drug abuse. For instance, the United States and the United Kingdom, which both contain relatively high-income inequality compared to other wealthy nations, also have a high prevalence of illicit drug use disorders and alcohol-related harms compared to their more egalitarian counterparts. The findings were consistent even after accounting for average incomes, which suggests that the gap between rich and poor correlates with higher substance abuse.

Studies that focus specifically on substance abuse also found similar findings. The extensive cross-national survey of adolescents conducted by the Health Behavior in School-aged Children Study examined alcohol use in 34 countries, each containing varied inequality levels. Elgar et al.'s analysis found that young adolescents (ages 11 and 13) in countries with high income inequality were more likely to consume alcohol and to have gotten drunk, compared to those in more equal societies (Elgar et al., 2005). Notably, this effect of inequality on youth drinking was observed even after controlling for individual family affluence and overall national wealth, indicating a contextual influence of the income gap itself. While adolescent drinking is

only one aspect of substance use, this finding echoes a piece of a broader theme that inequality at the societal level can permeate down to influence individual risk behaviors.

Focusing more on the United States, where both inequality and the opioid crisis are pronounced, researchers have explored similar links. Data from the Centers for Disease Control and Prevention (CDC) show that areas with greater income inequality tend to experience higher rates of drug overdose deaths (Centers for Disease Control and Prevention 2022). For example, a CDC analysis noted that U.S. counties with large income gaps had disproportionately high overdose mortality across most racial/ethnic groups (Johnson 2022). A potential contributor is that high inequality often coincides with pockets of concentrated disadvantage neighborhoods or counties, where poverty, unemployment, and low education levels are clustered. Such conditions can facilitate drug problems through multiple pathways: weaker economic opportunities, more disinvested social services, and potentially more illicit drug activity in the depressed area.

Adding nuance to this discussion is Karriker-Jaffe et al. (2013), who argue for a more nuanced approach that incorporates race-based dimensions of inequality. In their study, they find that between-race income disparities (specifically Black-White and Hispanic-White poverty ratios) are more predictive of alcohol consumption patterns and alcohol-related problems than Gini coefficients alone. Most notably, Black and Hispanic individuals experience a disproportionate burden of alcohol consequences and dependence in states with high racialized income inequality. These findings are particularly relevant as they suggest a “double jeopardy” effect, where racial minorities are both more exposed to structural inequality and more vulnerable to its negative health outcomes. Subsequently, the Gini coefficient may understate the severity of health disparities, and thus it is important to disaggregate the effects by racial composition as well.

Plenty of research has also investigated the specific mechanisms that link inequality to drug outcomes. A study by Yang et al. focused on opioid painkiller prescribing rates as an intermediate outcome reflecting community opioid exposure. The authors hypothesized that income inequality could lead to more opioid prescriptions, possibly because areas with high inequality have more social disintegration or unmet needs that doctors inadvertently address with medications. However, their results found no significant direct relationship between local Gini coefficients and prescribing rates. Instead, the effect of inequality is mediated through residential stability and social isolation. Therefore, high inequality was associated with less residential stability and greater social isolation among older adults, which then in turn became positively linked to opioid prescribing (Yang et al., 2021). These indirect pathways help explain how inequality may erode community cohesion and increase vulnerability to opioid overuse and add complexity to the argument that inequality directly drives addiction. In essence, they challenge oversimplified narratives of inequality addiction, where they emphasize how place-based social structures shape how inequality translates into health outcomes.

Not all studies find a large role for economic inequality or decline in driving the opioid epidemic, which adds even more nuance. Economist Christopher Ruhm investigated the extent to which the rise in drug mortality from 1999 to 2015 aligns with the "deaths of despair" narrative of declining economic conditions compared to other factors such as the drug supply. He found that counties experiencing relative economic decline did have somewhat higher growth in drug overdose mortality, but that the relationship was weak and largely explained by other confounding factors (Ruhm 2018). In his analysis, changes in medium-run indicators (such as job and income loss) only accounted for less than ten percent of the increase in opioid death rates during that period, once adjusted for other variables. This finding suggests that while inequality

and economic malaise may set the stage, the spikes in overdose deaths might be driven more by supply-side factors such as drug availability, distribution networks, and clinical practices. Other researchers have also pointed out the critical role of the drug environment, citing the oversupply of prescription opioids followed by the surge of illicit fentanyl as key causes of the mortality epidemic (Fischer et al., 2020). Nevertheless, it will be folly to assume that economic factors are not important. The consensus in the literature is that income inequality and related socioeconomic stressors are significant risk factors for addiction and overdose, even if they are not the sole triggers.

The Rural–Urban Divide in Addiction and Inequality

The literature reveals that there is also variation in impact across the rural-urban continuum. Research shows that non-metropolitan (rural) counties and metropolitan (urban) counties have different patterns when it comes to inequality (Thiede et al., 2020). Rural areas have relatively low within-county inequality, which is not too surprising as often everyone is similarly low-income. In contrast, urban areas tend to have larger inequality, as cities contain concentrated wealth alongside poverty. Interestingly, in some cases, there is more economic mobility within these urban areas. However, there is a key difference because the "scale of social stratification" may feel different between the two (Thiede et al., 2020). Specifically, in a rural community, everyone may know who the wealthy and poor are as the social interactions of each group are within close proximity, whereas in a city, the inequality may be more spatially segregated. As a result, these differences may influence how inequality impacts social cohesion and drug use. Indeed, as noted earlier, one study found inequality's association with opioid supply stronger in rural settings, suggesting that rural communities may be more sensitive to the

social disruptive effects of inequality (perhaps because there are fewer buffers or alternative support systems when the local economy falters).

Rural America has been the center of many conversations, being depicted as the ground zero for the prescription opioid misuse of the 2000s. Historical accounts and epidemiological data showcase support for this through the decline of rural Appalachia, which was among the first and hardest hit by opioid addiction. Possibly, this trend was due to the collapse of coal mining and manufacturing jobs, which has left a void of despair, which coincided with the aggressive distribution and marketing of prescription painkillers (Novack et al., 2020). A review by Novack et al. (2020) found that poor economic opportunity from the deindustrialization facilitated the opioid epidemic's spread in Appalachia. In these communities, prescription opioids became a culturally accepted form of relief, a medical solution to both the literal and psychic pain the populations were experiencing. This narrative expresses how rural contexts can amplify the synergy between economic despair and drug availability, producing a severe addiction crisis.

By contrast, urban areas have their own distinct dynamics. Urban centers often have more established illicit drug markets and distribution networks, making them more widespread. However, urban areas also typically have greater healthcare infrastructure (e.g., more hospitals, clinics, and addiction treatment providers), giving a tradeoff. While rural communities may have had lower population exposure, they have higher vulnerability due to economic conditions and limited resources, meaning that urban communities, despite having larger exposed populations, have better access to interventions (albeit with many underserved groups still). Thus, one of the major disparities is in access to addiction treatment and healthcare services. Franz et al. (2024) examined the availability of hospital-based opioid use disorder services (like screening, addiction

consult teams, and medications for opioid use disorder). (Franz et al., 2024). They found that although there were no significant rural-urban differences in the availability of substance use disorder screening in emergency departments, rural hospitals had significantly lower odds of offering addiction consult services in emergency departments compared to metropolitan hospitals. Moreover, medications for opioid use disorder were much lower in rural hospitals. This lack of services in rural communities means that individuals struggling with addiction may not get help at critical moments. The authors conclude that rural communities face unique needs in the opioid epidemic, and yet their hospitals and clinics are often ill-equipped to implement the gold standard interventions for addiction.

Beyond the medical treatments, there are social determinants that differ between rural and urban areas. Rural populations are older on average and have higher rates of poverty. These factors can compound the impact of deaths of despair. Furthermore, rural drug users may be more isolated both socially and geographically, making it harder for them to reach resources closest to them, and raising the risk of using alone and suffering a fatal overdose without intervention. Alternatively, there may be some protective factors stronger in certain rural communities, such as tight-knit social networks and faith-based support, though these can be eroded by high inequality and out-migration.

In essence, geography matters in how inequality translates to drug addiction and overdose risk. Rural areas have been faced with the burdens of economic decline and inadequate healthcare infrastructure, exacerbating deaths of despair. However, urban areas have not been spared, as they also have inequality and despair in certain neighborhoods, but they often have more resources that can mitigate the harm. This study's focus on county population as an

interaction variable (proxying rural vs urban context) is motivated by this understanding that the rural-urban divide is a key piece of the puzzle.

Gaps and Rationale for the Present Study

Despite the rich literature reviewed above, there are still plenty of important gaps worthy of exploring. Specifically, there has not been much focus on the intersection of economic and geographic factors. The research has shown that inequality may have stronger effects in rural areas through opioid prescription (Yang et al., 2021). However, this insight has not been fully explored for outcomes like overdose death rates or other methods of measuring addiction prevalence. Most analyses look at inequality's influence at a broad level or focus on rural vs urban differences in outcomes, but rarely both simultaneously. This leaves a gap to pinpoint if a high-Gini County of a small population experiences a different addiction outcome trajectory than that of a large population. By understanding this interaction, one can be more informed on proper interventions, whether they should be place-specific, and if inequality is a component that should be taken into account.

Moreover, there are gaps with the limited local-level analysis. Many of the cross-sectional studies referenced above use state-level or national-level inequality measures. Consequently, differences between Gini coefficients of two counties in the same state get ignored along with their social/local economic dynamics. This literature would benefit tremendously from more county-level analyses in linking inequality or other economic measures to substance-related outcomes. This study attempts to fill this gap by using county-level Gini coefficients and drug mortality rates. Additionally, using county population as an interaction variable effectively allows us to stratify by rural vs urban character (since population correlates with urbanization).

Another interesting component about substance abuse that can get overlooked is where the focus lies with particular substances. While opioids rightfully deserve their focus given the crisis, it is worth noting that inequality's impact may extend to other substances such as alcohol. Studies tend to focus on drugs or alcohol alone, as each contains its own respective problems. Therefore, in this study, a comparison will be made between crude rates of alcohol and drugs to parse out differences between the two.

In light of these gaps, the current study is warranted. It will specifically address the need to jointly consider income inequality and rural-urban context in explaining drug addiction outcomes, while also controlling for many demographic and economic indicators. The key component to test this primary hypothesis is through an interaction variable between inequality and population. This variable will test whether the impact of inequality on overdose rates is moderated by community type, building upon the prior research that rural communities may be especially vulnerable to the effects of inequality (Yang et al., 2021). If significant, this finding will suggest that policies to mitigate addiction should not only aim to reduce inequality generally but also target the unique challenges of rural areas.

Furthermore, this study will help reconcile the narrative of "deaths of despair" with more recent critiques. If we find strong associations between inequality and drug mortality, it reinforces the Case and Deaton framework that structural economic factors underlie the crisis. If the association is weak or mostly indirect, it aligns with Ruhm's findings that other factors dominate, thereby shifting focus to those such as drug supply. Either result is informative, and strong evidence in either direction will underscore the importance of specific interventions and whether they should be economically focused.

In conclusion, the literature to date paints a compelling picture of economic factors being linked to higher drug addiction and overdose rates, and that rural communities may face distinct vulnerabilities in this regard. However, important nuances and mechanisms remain under-explored, so this present thesis seeks to contribute by thoroughly examining how the Gini coefficient relates to county-level drug mortality, and whether this relationship differs by rural versus urban setting, proxied by population size. In doing so, the gap in understanding with the interplay between economic inequality and geography in fueling the ongoing crisis of addiction can take a critical step to better inform effective and targeted solutions to reduce "death of despair" in America's diverse communities.

Methods

Data Sources and Collection

County-level data for drug related mortality was retrieved from the Centers for Disease Control and Prevention (CDC) using the Wide-ranging Online Data for Epidemiological Research (CDC WONDER) platform. Given the unique impact of the COVID-19 pandemic on mortality trends in 2020, I used a five-year average (2016-2020) of crude overdose rates to mitigate annual volatility and improve the stability of the dependent variable. Two separate datasets (details listed below in the summary statistics section) were created, one for drug and one for alcohol mortality to compare variations between the two.

Socioeconomic and demographic variables were obtained through the U.S. Census Bureau's American Community Survey (ACS), using the five-year estimates from 2020 (i.e., 2016-2020), ensuring temporal consistency. Variables included: the Gini coefficient (a measure of income inequality from 0-1, which was rescaled to 0-100 for interpretability), median household income, percentage of non-Hispanic white residents, unemployment rate, and labor force participation rate.

To ensure data quality, counties with missing or statistically unreliable values, such as being flagged as unreliable or missing by either the CDC WONDER or ACS, was excluded. Subsequently, this is how there came two different datasets, with one containing counties with complete data on drug-related mortality and socioeconomic indicators and another for alcohol-related mortality and the same socioeconomic variables.

Data Integration and Preparation

Raw data from the CDC and ACS were first organized and merged in Microsoft Excel using county-level Federal Information Processing Standards (FIPS) codes. Discrepancies in formatting were addressed by standardizing county codes (i.e., padding with leading zeros to ensure five-digit consistency). Once merged, each data set was exported to a .csv format and imported into Stata for further cleaning. Within Stata, variables were verified for correct storage types and converted as needed (e.g., STR to FLOAT). Once successfully cleaned and organized, the data was transported into a .dta file for simple future access and statistical analysis.

Empirical Strategy and Statistical Modeling

The primary estimation method was Ordinary Least Squares (OLS) regression, applied separately to the drug and alcohol datasets. For each model, the dependent variable was the respective county-level crude death rate (drug-induced or alcohol-induced). Independent variables were added in a nested modeling structure, allowing the analysis to assess how the relationship between structure factors and mortality outcomes change with the inclusion of demographic and labor market controls. The key model variations tested are as followed: the linear relationship between inequality and overdose mortality; nonlinear specifications via inclusions of a squared Gini term, and in some cases, interaction terms (e.g., Gini x population) to evaluate moderation effects.

Predicted values were later visualized through marginal effects plots across the range of key variables, holding others constant. For nonlinear models, the turning point of the relationship was calculated using the vertex formula for quadratic equations, helping identity the level of inequality or labor force participation associated with minimum predicted mortality. All statistical analysis and graphing were conducted using Stata and all resulting figures and tables are presented in the Results and Appendix.

Results

Summary Statistics

Table 1: Descriptive Statistics of Drug Crude Rate Data Set

	Mean	Std. Dev.	Min	Max
Population	288,302	410,866	16,916	4,678,621
Drug Crude Rate	33.3	18.69	4.26	171.40
Gini	0.45	0.03	0.35	0.60
Median Income	63,402.17	17388.32	26,072	147,111
Labor Force Participation Rate	61.93	6.14	28.7	78.9
Unemployment Rate	5.03	1.18	2.4	11.32
Percent of White Population	79.18	14.9	16.09	97.66

Total Counties Used: N = 555

Table 1 presents summary statistics for the drug crude rate dataset, which includes 555 U.S. counties from 2016 to 2020. On average, counties reported 33.3 drug-induced deaths per 100,000 residents (SD = 18.69), with rates ranging from 4.26 to 171.4. The average Gini coefficient—scaled from 0 to 1—was 0.45, indicating moderate income inequality, with a range from 0.35 to 0.60.

Counties in the sample had an average labor force participation rate of 61.93% (SD = 6.14), and an unemployment rate of 5.03%. The median household income averaged $63,402, ranging from approximately $26,072 to $147,111. The average percentage of non-Hispanic white residents was 79.18%, though this varied widely from 16.09% to 97.66%, reflecting considerable demographic diversity.

Table 2: Descriptive Statistics of Alcohol Crude Rate Data Set

	Mean	Std. Dev.	Min	Max
Population	450,866	694,272	10,313	10,100,000
Alcohol Crude Rate	17.39	13.66	3.5	193.9
Gini	0.45	0.03	0.38	0.60
Median Income	66,165.35	17,152.68	24,102	147,111
Labor Force Participation Rate	62.94	6.18	22.60	76.40
Unemployment Rate	5.00	1.34	2.40	17.40
Percent of White Population	74.41	15.89	9.35	96.01

Total Counties Used: N = 547

Table 2 summarizes the alcohol crude rate dataset, which also includes 555 counties. The average alcohol-induced crude mortality rate was notably lower at 17.39 per 100,000 (SD = 13.66), although maximum values reached nearly 194 deaths per 100,000. Similar to the drug dataset, these counties had an average Gini coefficient of 0.45, with comparable income distributions.

Median household income was slightly higher in this sample ($66,165), and the average labor force participation rate was also marginally greater (62.94%). Demographically, these counties were slightly less white on average (74.41%) than those in the drug dataset, though they similarly exhibited wide variation.

These statistics provide a contextual overview of the county-level structural conditions across both datasets and form the basis for the regression analyses that follow.

Regression Results: Drug Crude Rate Mortality

Table 3: Drug Crude Rate Models

	(1) Baseline	(2) Linear Full Model	(3) Nonlinear Full Model
Gini	0.682**	-0.040	-7.714*
	(0.257)	(0.311)	(3.709)
Gini X Population		7.61e-07	
		(5.01e-07)	
Labor Force Participation Rate		-0.990***	-9.115***
		(0.195)	(1.158)
Population		-0.0000406	-4.95e-06*
		(0.0000242)	(1.91e-06)
Percent White		0.100	0.141**
		(0.055)	(0.053)
Median Income		-0.0000414	-0.0001571**
		(0.0000557)	(0.0000546)
Unemployment Rate		2.822**	2.848**
		(0.807)	(0.769)
$Gini^2$			0.088*
			(0.041)
Labor Force Participation $Rate^2$			0.07***
			(0.01)
Constant	2.634	67.120**	481.157***
	(11.539)	(22.847)	(90.401)
Adjusted R^2	0.0108	0.303	0.3673
BIC	4816.099	4652.687	4605.692

Standard Errors in Parenthesis
N = 555
** p < 0.05, ** p < 0.01, *** p < 0.001*

Table 3 presents three progressively specified ordinary least squares (OLS) regression models estimating the relationship between structural socioeconomic conditions and county-level drug-induced crude death rates. The models begin with a simple baseline specification that only incorporates the Gini coefficient (Model 1), then incorporates a full set of demographic and economic controls linearly (Model 2), and then finally expands to account for nonlinear dynamics in both inequality and labor force participation (Model 3).

In Model 1, the Gini coefficient (scaled as a percentage) from 0 to 100) is the sole predictor of drug crude mortality. The results indicate a statistically significant and positive relationship ($B = 0.682$, $p<0.01$), suggesting that a one-point increase in the Gini index is associated with an estimated 0.68 increase in the drug induced crude death rate per 100,000 population. While this model is extremely limited in scope, the significance of Gini in isolation gives support to the base hypothesis of income inequality being associated with greater vulnerability to drug-related mortality at the county level. However, as seen with the tiny R^2 ($R^2 = 0.0108$), there is a high likelihood of omitted variable bias through unobserved confounders.

In the Full Linear Specification of the model, I introduce theoretically relevant structural controls and my interaction term Gini x Population that tests whether the inequality-mortality relationship is moderated by county population size. After introducing these controls, the coefficient on Gini becomes statistically insignificant and near zero ($B = -0.040$, $p > 0.10$), suggesting that the previously significant association in Model 1may have been a reflection of confounding structural characteristics rather than a direct effect of inequality alone. Meanwhile, labor force participation rate emerges as a robust predictor with counties containing higher participation rates experiencing significantly lower crude death rates ($B = -0.990$, $p < 0.001$).

Similarly, unemployment rate is also significant but with a positive association ($B = 2.8222$, $p < 0.01$), which supports the idea that economic disharmony can contribute to substance-related harm. Importantly, the interaction term was insignificant, showing that the impact of inequality is not moderated through population, which fails to support the hypothesis that there is an urban rural divide.

Model 3 builds upon the prior specification by incorporating nonlinear terms for both inequality and labor force participation, reflecting the idea that these relationships may instead follow a U-shaped or diminishing return pattern, rather that those of simple linear dynamics. An interaction term between income inequality and population size (Gini × population) was initially included in model 3 to test whether the relationship between inequality and drug mortality varied by county size. However, this specification produced extreme multicollinearity (mean VIF > 200), which destabilized coefficient estimates and rendered key terms statistically insignificant. After removing the interaction, multicollinearity was substantially reduced, and both the linear and nonlinear inequality terms emerged as statistically significant predictors. The final model excludes the interaction in favor of a more interpretable and statistically reliable specification. The results support the nonlinear hypothesis, with the Gini coefficient becoming significant again ($B = -7.714$, $p < 0.05$) along with its squared counterpart ($B = 0.088$, $p < 0.05$). This relationship implies a U-shaped relationship between inequality and drug related mortality, where counties with either very low or very high-income inequality both tend to have higher crude drug death rates, whereas those in the middle experience the lowest level mortality. This same pattern happened with labor force participation rate. These findings reflect the complex social dynamics underpinning community vulnerability.

Other covariates also remain significant in expected directions. Interestingly, there is a positive effect of percent white that aligns with the broader demographic patterns observed in the opioid crisis, which has disproportionate impacted non-Hispanic white populations in rural and suburban counties.

Overall, the full nonlinear specification model is the strongest performing both substantively and statistically. The adjusted R^2 rises to 0.3673, indicating that the model explains over 36% of the variation in county-level drug crude death rates, which is a considerable improvement over the first two models. Additionally, Model 3 yields the lowest BIC (4605.692), confirming it as the best-fitting model among the three. These indicators combined with the theoretical interpretability of the nonlinear terms, support the selection of Model 3 as the final preferred model.

Figure 1

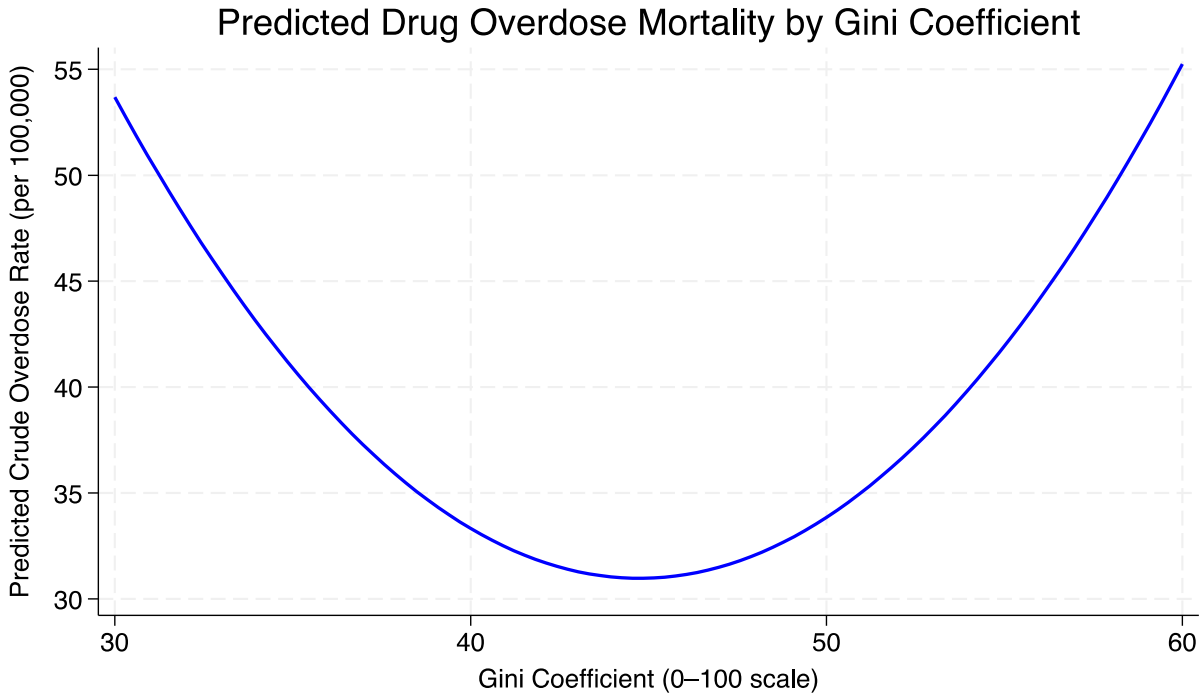

Figure 1 visualizes the predicted relationship between county-level income inequality and drug-induced crude death rates as seen in Model 3. Initially, increases in the Gini are associated with lower drug crude rates, but after a certain threshold, further increases lead to rising drug crude rates. The minimum of the curve occurs at a Gini index of approximately 44.9 on a 0-100 scale (or 0.449 from the typical index) and was calculated with the vertex formula for a quadratic function: $X^* = -B_1/2B_2$, where B_1 and B_2 are the linear and squared Gini coefficients from Model 3. As a result, counties from either above or below the threshold experience higher rates of overdose mortality.

Figure 2

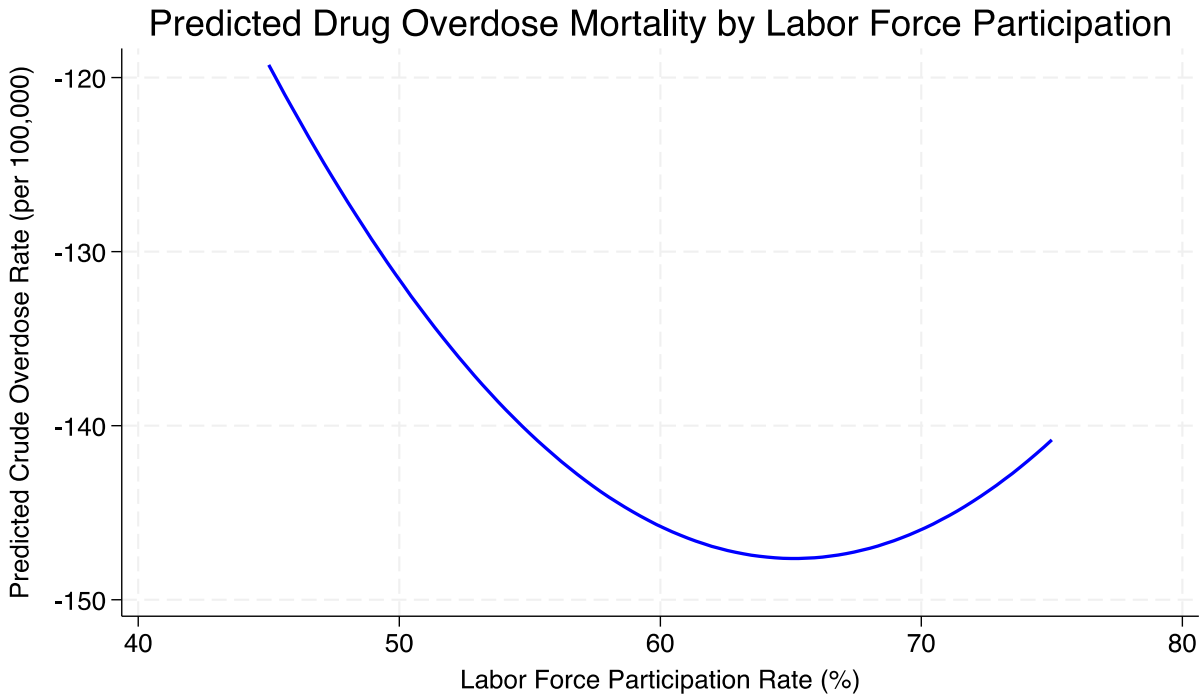

Figure 2 does the same as figure 1, but this time for labor force participation rate. Based on the figure, there is a clear U-shaped pattern, where mortality rates decline with increasing labor force participation up to a certain point, after which they begin to rise. The minimum point

calculated through the vertex formula indicates that there is a minimum at approximately 65.1% labor force participation. Notably, compared to the U-shaped curve in Figure 1, the right side of the U-shape in Figure 2 rises much more gradually.

Regression Results: Alcohol Crude Rate Mortality

Table 4: Alcohol Crude Rate Models

	(1) Baseline	(2) Linear Full Model	(3) Nonlinear Full Model
Gini	0.304 (0.189)	-0.433 (0.230)	-5.124 (3.107)
Gini X Population		3.84e-07 (2.96e-07)	
Labor Force Participation Rate		-0.884*** (0.115)	-3.57*** (0.757
Population		-0.000214 (-0.0000145)	-2.59e-06** (7.96e-07)
Percent White		-0.100* (0.039)	-0.054 (0.038)
Median Income		-0.0001131** (0.0000379)	-0.0001618*** (0.0000381)
Unemployment Rate		-0.133 (0.484)	0.243 (0.485)
$Gini^2$			00535 (0.034)
Labor Force Participation $Rate^2$			0.024*** (0.007)
Constant	3.581 (8.605)	109.67*** (16.245)	284.14*** (72.67)
Adjusted R^2	0.0029	0.270	0.2885
BIC	4421.246	4282.992	4273.732

Standard Errors in Parenthesis
N = 547
** p < 0.05, ** p < 0.01, *** p < 0.001*

Table 4 conducts the same analysis and process of thinking as table 3, but it instead works with the alcohol-induced crude death rate dataset. As with the drug mortality models, Model 1 includes only the Gini coefficient, Model 2 introduced full linear controls, and Model 3 incorporates nonlinear terms for both inequality and labor force participation.

In contrast to the drug dataset, the Gini coefficient does not exhibit consistent statistical significance across specifications, including the baseline specification. However, labor force participation rate emerges as a consistently strong predictor, containing a negative and highly significant value in both Model 2 ($B = -0.884$, $p < 0.0001$) and Model 3 ($B = -3.57$, $p < 0.001$), while the squared term indicates a significant nonlinear component similar to the drug models ($B = 0.024$, $p < 0.001$). Unemployment rate and percent white do not appear to be significant predictors unlike their relevance in the drug models. Median income retains a modest but consistent negative association across the last two models.

Model fit statistics show a similar pattern to the drug regressions: adjusted R^2 improves from 0.0029 in the baseline to 0.270 in the full linear model, and slightly more to 0.2885 in the nonlinear specification. The Bayesian Information Criterion (BIC) decreases across models, indicating improved fit, though not as markedly as in the drug mortality case.

Figure 3

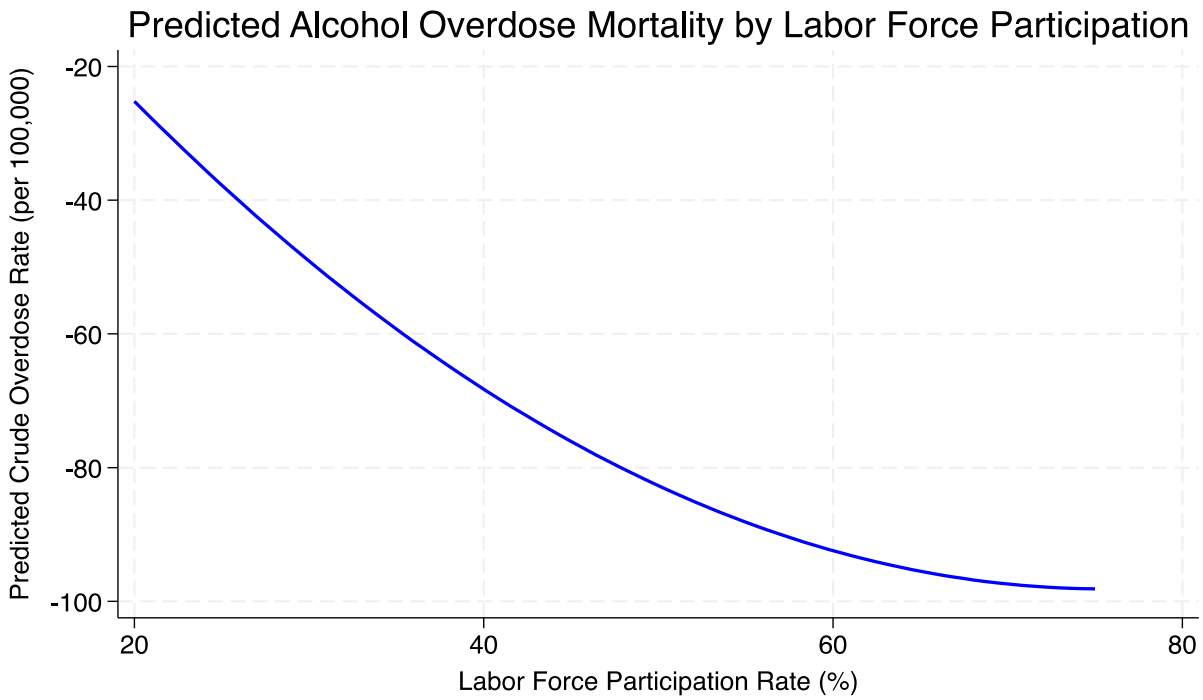

Figure 3 illustrates the predicted relationship between labor force participation rate and county-level alcohol-induced crude death rates based on the nonlinear specification from Model 3. The curve demonstrates a consistently negative, concave shape, reflecting that increases in labor force participation are associated with lower predicted alcohol mortality. Unlike the drug mortality model, this relationship does not form a U-shape. While the squared term for this was significant, the graph suggests that it has a diminishing marginal returns effect, rather than a complete reverse in direction.

Discussion

This study started off with the goal to examine the effects of income inequality on drug addiction by examining county-level Gini coefficients to their respective county-level crude death rates. Specifically, I was interested if these relationships varied across the rural-urban

continuum. Several key findings emerged that align with the existing literature while also offering new insights.

Drug Crude Rates and the White Population

One of the most striking results is the consistent significance of the *percent white* population in predicting drug crude death rates. This finding aligns with the literature on Deaths of Despair as proposed by Case and Deaton, which have documented the disproportionally high rates of substance-related mortality among working-class white populations. This demographic has faced socioeconomic stagnation, erosion of traditional employment opportunities, and rising social disconnection, which are all factors that contribute to the noted substance use disorders. The significance of *percent white* in my model supports this idea that these broader structural and cultural conditions continue to shape vulnerability to drug mortality in a way that warrants targeted policy attention.

Nonlinear Effects of Income Inequality

A major finding to this econometric analysis concerns the U-shaped relationship seen between the Gini coefficient and drug crude death rates, as shown in the quadratic model. This counterintuitive pattern may reflect the dynamics where extremely equal counties may be uniformly low-income with minimal economic activity, resulting in more conditions that affect drug use. Moreover, on the extreme end of inequality there corresponds sharp increases in drug mortality, suggesting that excessive economic disparity generates conditions that can weaken community cohesion, becoming a mechanism for social stress and death of despair frameworks. Interestingly, this sort of dynamic is not replicated in the alcohol crude death rate dataset being insignificant, which could possibly be explained by the social normalization and availability of alcohol across economic and demographic groups. Unlike illicit drugs, alcohol is legally and

widely accessible, and its consumption may not be as closely tied to inequality-driven stress or social isolation.

Interpreting Labor Force Participation Trends

Arguably, the most interesting findings in this study involves labor force participation rate. What was once, simply a control, generated key findings worth discussing and becoming the center of conversation. This factor revealed a nonlinear relationship with drug crude rates. However, unlike the Gini coefficient, the shape is not strictly U-shaped, but rather more nuanced. In general, higher labor force participation is associated with lower crude rates. A possible explanation for that is that employment is often a protective factor against drug use. Nevertheless, at more extreme levels of labor force participation the trend appears to reverse slightly. What may be occurring here is an income effect, where affluent or highly employed counties contain individuals who have more disposable income, which can paradoxically increase access to substances. This is consistent with the stereotype of the high-functioning, high0income drug user, such as a business executive who engages in substance use despite financial security. This interpretation suggests that labor force participation is not unambiguously protective and may interact with other economic or cultural factors that modulate drug access and consumption norms. The relationship with alcohol crude rates was also interesting, being highly significant and nonlinear, yet following a diminishing returns pattern rather than a truly complete U-shape. This suggests that while employment may offer general protective benefits (such as routine, social integration, and income), these benefits may plateau, and other factors like stress or drinking culture in certain occupations may limit further reductions in alcohol-related harm.

Overall, these findings explain how truly complicated the relationship between structural variables and substance use outcomes really are. They suggest that inequality and economic participation exert distinct and non-linear effects depending on the substance in question and the demographic context. These results support many of the hypotheses on the issue of drug addiction, and they also highlight how interventions must consider the complex dynamics, especially if they are focused on going the economic route.

Conclusion

Limitations

First, the analysis is fundamentally correlational and relies on county-level aggregate data, which limits causal inference. Although patterns at the county level can reveal important trends, they may obscure significant within-county heterogeneity. For example, differences in access to healthcare, housing stability, or social services across neighborhoods. Additionally, economic analyses cannot speak to individual-level mechanisms, meaning that the study cannot directly determine how personal experiences of inequality, employment, or racialized disadvantage influence drug or alcohol use.

Second, the study uses five-year averages (2016–2020) for both mortality and socioeconomic indicators. This approach smooths year-to-year volatility and increases the reliability of estimates, particularly for smaller or rural counties where annual fluctuations might otherwise distort patterns. However, this smoothing also has trade-offs: it may dilute recent trends, especially those affected by COVID-19-related disruptions in healthcare access, substance use patterns, and labor market dynamics. The averaging may mask important short-

term shocks or rapid changes in community vulnerability that occurred during or after the onset of the pandemic.

Third, while the regression models incorporate a set of theoretically grounded control variables to reduce omitted variable bias, there are still other potential confounders not included in the model. These may include factors such as availability and quality of healthcare services, local criminal justice practices, prevalence of mental health disorders, housing insecurity, or regional cultural attitudes toward substance use. The exclusion of such variables may introduce unobserved heterogeneity that affects the observed associations.

An additional limitation came from data availability and quality, particularly in rural counties. County-level socioeconomic and mortality data are often more difficult to obtain for sparsely populated or under-resourced regions, which can lead to missing data, imprecise estimates, or measurement error, especially for variables like crude death rates that are sensitive to small population sizes. This issue can introduce bias or reduce the reliability of estimated associations in rural areas. Additionally, the study first intended to incorporate the Rural-Urban Continuum Code (RUCC) to more precisely capture rurality and differentiate between degrees of urbanization, but because of limited and inconsistent coverage across key variables, the RUCC could not be systematically applied across all observations. As a result, population size was used as a proxy for rurality, which, while useful, may not fully reflect the complex socioeconomic and geographic characteristics that distinguish rural from urban communities.

Lastly, the measurement of key predictors such as income inequality via the Gini coefficient may not fully capture the multidimensional nature of economic stratification or social fragmentation. For instance, the Gini coefficient does not reflect racialized income disparities,

wealth inequality, or intergenerational poverty, all of which may shape substance use outcomes in ways distinct from income concentration alone.

Future Studies

Based on the limitations of this study, there are many ways to go for future studies to better understand drug addiction and how economic variables may impact this relationship. For instance, future studies could enhance the geographic specificity of rural-urban classifications to better differentiates between degrees of remoteness and metropolitan localities. Future research could explore temporal dynamics, particularly in light of the COVID-19 pandemic. Examining pre and post pandemic trends can provide plenty of insight on this issue. Finally, future studies could expand the scope by including separate measures for drug addiction that may be more direct and less of a proxy to the issue.

Final Words

This thesis has contributed to the growing body of research surrounding drug-related mortality by highlighting its complexity regarding the relationship to economic variables. By using crude death rates as a proxy for addiction and incorporation interaction terms to explore rural-urban differences, the analysis does not reveal an interaction effect there. However, the analysis did reveal a nonlinear relationship in regard to Gini and labor force participation. The findings show how important it is that policy solutions remain nuanced in addressing the crisis of substance abuse. As the U.S. continues to battle rising overdose death rates, especially in rural communities, studies like this one offer a foundation for more targeted and equitable public health strategies.

References

Baltagi, B. H., & Griffin, J. M. (2001). The econometrics of rational addiction: the case of cigarettes. Journal of Business & Economic Statistics, 19(4), 449-454.

Case, A., & Deaton, A. (2021). Deaths of Despair and the Future of Capitalism.

Centers for Disease Control and Prevention. (2022, July 19). Drug overdose deaths rise, Disparities Widen. Drug Overdose Deaths Rise, Disparities Widen Differences Grew by Race, Ethnicity, and Other Factors. https://www.cdc.gov/vitalsigns/overdose-death-disparities/index.html

Elgar, F. J., Roberts, C., Parry-Langdon, N., & Boyce, W. (2005). Income inequality and alcohol use: a multilevel analysis of drinking and drunkenness in adolescents in 34 countries. The European Journal of Public Health, 15(3), 245-250.

Fischer, B., Pang, M., & Jones, W. (2020). The opioid mortality epidemic in North America: do we understand the supply side dynamics of this unprecedented crisis? Substance abuse treatment, prevention, and policy, 15, 1-8.

Franz, B., Cronin, C. E., Lindenfeld, Z., Pagan, J. A., Lai, A. Y., Krawczyk, N., ... & Chang, J. E. (2024). Rural-urban disparities in the availability of hospital-based screening, medications for opioid use disorder, and addiction consult services. Journal of Substance Use and Addiction Treatment, 160, 209280.

Gould, R. W., Czoty, P. W., Porrino, L. J., & Nader, M. A. (2017). Social status in monkeys: effects of social confrontation on brain function and cocaine self-administration. Neuropsychopharmacology, 42(5), 1093-1102.

Heyman, G. M., Ryu, E., & Brownell, H. (2024). Evidence that intergenerational income mobility is the strongest predictor of drug overdose deaths in US Midwest counties. International Journal of Drug Policy, 132, 104558.

Johnson, S. R. (2022, July 19). Study: Overdose deaths increased with income inequality. https://www.usnews.com/news/health-news/articles/2022-07-19/overdose-deaths-among-blacks-hispanics-increased-with-income-inequality

Karriker-Jaffe, K. J., Roberts, S. C., & Bond, J. (2013). Income inequality, alcohol use, and alcohol-related problems. American journal of public health, 103(4), 649–656. https://doi.org/10.2105/AJPH.2012.300882

Latimore, A., Hinojosa, A. O., Kestner, L., & Patel, P. (2021, November 19). Exploring urban-rural disparities in accessing treatment for opioid use disorder | american institutes for Research. Exploring Urban-Rural Disparities in Accessing Treatment for Opioid Use Disorder. https://www.air.org/resource/field/exploring-urban-rural-disparities-accessing-treatment-opioid-use-disorder

Lister, J. J., Weaver, A., Ellis, J. D., Himle, J. A., & Ledgerwood, D. M. (2020). A systematic review of rural-specific barriers to medication treatment for opioid use disorder in the United States. The American journal of drug and alcohol abuse, 46(3), 273–288. https://doi.org/10.1080/00952990.2019.1694536

Lundborg P. (2002). Young people and alcohol: an econometric analysis. Addiction (Abingdon, England), 97(12), 1573–1582. https://doi.org/10.1046/j.1360-0443.2002.00231.x

Marks, C., Abramovitz, D., Donnelly, C. A., Carrasco-Escobar, G., Carrasco-Hernández, R., Ciccarone, D., ... & Bórquez, A. (2021). Identifying counties at risk of high overdose

mortality burden during the emerging fentanyl epidemic in the USA: a predictive statistical modelling study. The Lancet Public Health, 6(10), e720-e728.

Nagelhout, G. E., Hummel, K., de Goeij, M. C. M., de Vries, H., Kaner, E., & Lemmens, P. (2017). How economic recessions and unemployment affect illegal drug use: A systematic realist literature review. The International journal on drug policy, 44, 69–83. https://doi.org/10.1016/j.drugpo.2017.03.013

Novack, L. M., Carrasco, K. G., Tyler, K. A., Dombrowski, K., & Habecker, P. (2020). Injection opioid and injection methamphetamine use in the rural United States: A systematic review and network analysis. Journal of Drug Issues, 50(2), 127-141.

Rhodes, J. E., & Jason, L. A. (1990). A social stress model of substance abuse. Journal of Consulting and Clinical Psychology, 58(4), 395–401. https://doi.org/10.1037/0022-006X.58.4.395

Ruhm, C. J. (with National Bureau of Economic Research). (2018). Deaths of Despair or Drug Problems? National Bureau of Economic Research.

Scutchfield, F. D., & Keck, C. W. (2017). Deaths of Despair: Why? What to Do?. American journal of public health, 107(10), 1564–1565. https://doi.org/10.2105/AJPH.2017.303992

Thiede, B. C., Butler, J. L., Brown, D. L., & Jensen, L. (2020). Income inequality across the rural-urban continuum in the United States, 1970–2016. Rural Sociology, 85(4), 899-937.

Wilkinson, R. G., & Pickett, K. E. (2009). Income inequality and social dysfunction. Annual review of sociology, 35(1), 493-511.

Yang, T. C., Kim, S., & Shoff, C. (2021). Income inequality and opioid prescribing rates: exploring rural/urban differences in pathways via residential stability and social isolation. Rural sociology, 86(1), 26-49.

Appendix

Appendix A: STATA Code

```
*****Honors Thesis******
**By Jonathan Flores**
clear all
set more off

//use dataset with just drug crude rate
use "/Users/jonathanflores/Desktop/Honors Thesis/DrugData.dta", clear

//Inspect Variables
describe
summarize


************GINI Focused Nested Model Structure Drug Only Crude
Rates************

//Model0 Gini Only
gen gini_pct = gini * 100
regress drugcruderate gini_pct


//Model1 Addiciton, Inequality, Population, White Demographic*****
regress drugcruderate gini_pct population percentwhite

//Model2 Addiciton, Inequality, Population, Median Income, Unemployment Rate, Labor
Force Participation Rate*****
regress drugcruderate gini_pct population percentwhite medianincome unemploymentrate
laborforceprate



//Model3 Addiciton, Inequality, Population, Median Income, Unemployment Rate, Labor
Force Participation Rate, Interaction of Gini & Population*****

gen gini_population = gini_pct * population
regress drugcruderate gini_pct population percentwhite medianincome unemploymentrate
laborforceprate gini_population

//Model4 Nonlinear checks
gen gini2 = gini_pct^2
regress drugcruderate gini_pct population percentwhite medianincome unemploymentrate
laborforceprate gini2 gini_population
```

```
estat ic

range gini_range 30 60 100
gen gini_range2 = gini_range^2

gen y_hat = _b[_cons] + _b[gini_pct]*gini_range + _b[gini2]*gini_range2 ///
   + _b[population]*288301.6 ///
   + _b[percentwhite]*79.18373 ///
   + _b[medianincome]*63402.17 ///
   + _b[unemploymentrate]*5.033009 ///
   + _b[laborforceprate]*61.93297



twoway (line y_hat gini_range, lcolor(blue) lwidth(medthick)) ///
      , title("Predicted Drug Overdose Mortality by Gini Coefficient") ///
        xtitle("Gini Coefficient (0–100 scale)") ///
        ytitle("Predicted Crude Overdose Rate")



************* Labor Force Participation-Focused Model Structure *************

// Model 0: Labor Force Participation Only (baseline)
regress drugcruderate laborforceprate

// Model 1: Add population scale and demographic structure
regress drugcruderate laborforceprate population percentwhite

// Model 2: Add economic structure (income, unemployment, Gini)
regress drugcruderate laborforceprate population percentwhite medianincome
unemploymentrate gini_pct

// Model 3: Add interaction term (laborforceprate × population)
gen labor_population = laborforceprate * population
regress drugcruderate laborforceprate population percentwhite medianincome
unemploymentrate gini_pct labor_population

// Model 4: Add nonlinearity check (squared laborforceprate)
gen laborforceprate2 = laborforceprate^2
regress drugcruderate laborforceprate population percentwhite medianincome
unemploymentrate gini_pct laborforceprate2 labor_population
estat ic

range lfp_range 45 75 100
gen lfp_range2 = lfp_range^2
```

```
gen y_hat_lfp = _b[_cons] ///
   + _b[laborforceprate]*lfp_range ///
   + _b[laborforceprate2]*lfp_range2 ///
   + _b[gini_pct]*44.85809 ///
   + _b[population]* 288301.6 ///
   + _b[percentwhite]* 79.18373 ///
   + _b[medianincome]* 63402.17 ///
   + _b[unemploymentrate]* 5.033009

twoway (line y_hat_lfp lfp_range, lcolor(blue) lwidth(medthick)) ///
      , title("Predicted Drug Overdose Mortality by Labor Force Participation") ///
        xtitle("Labor Force Participation Rate (%)") ///
        ytitle("Predicted Crude Overdose Rate")


gen labor_unemployment = laborforceprate * unemploymentrate

// Model 6: BEST
regress drugcruderate laborforceprate population percentwhite medianincome
unemploymentrate gini_pct gini2 laborforceprate2 labor_population
estat ic

************GINI Focused Nested Model Structure Alcohol Only Crude
Rates************

//use dataset with just drug crude rate
use "/Users/jonathanflores/Desktop/Honors Thesis/AlcoholData.dta", clear

//Inspect Variables
describe
summarize

//model 0
gen gini_pct = gini * 100
regress alccruderate gini_pct
estat ic

//Model3
gen gini_population = gini_pct * population
regress alccruderate gini_pct population percentwhite medianincome unemploymentrate
laborforceprate gini_population
estat ic


// Model 6: BEST
```

```
gen gini2 = gini_pct^2
gen laborforceprate2 = laborforceprate^2
regress alccruderate laborforceprate population percentwhite medianincome
unemploymentrate gini_pct gini2 laborforceprate2
estat ic


range lfp_range 20 75 80
gen lfp_range2 = lfp_range^2

gen y_hat_lfp = _b[_cons] ///
   + _b[laborforceprate]*lfp_range ///
   + _b[laborforceprate2]*lfp_range2 ///
   + _b[gini_pct]*45.41833 ///
   + _b[population]* 450866.1 ///
   + _b[percentwhite]* 74.40692 ///
   + _b[medianincome]*  66165.35 ///
   + _b[unemploymentrate]* 5.003656

twoway (line y_hat_lfp lfp_range, lcolor(blue) lwidth(medthick)) ///
      , title("Predicted Alcohol Overdose Mortality by Labor Force Participation") ///
        xtitle("Labor Force Participation Rate (%)") ///
        ytitle("Predicted Crude Overdose Rate (per 100,000)")
```